

11.2 The F statistic as a comparison of projections

The formula Eq. 11.1.1 mentions two residual sums of squares, but the test is really about three subspaces and the way the data vector sits among them. Seen this way, the F statistic is a function of a single angle, its rejection region is a cone, and its null distribution survives far beyond normal errors. The same view gives a canonical form in which every linear hypothesis looks the same. [Section 11.4](#) needs that form to prove that the F test is optimal.

11.2.1 Three orthogonal pieces

By Theorem 6.5.1, nested model spaces split $\mathbb{R}^n$ into three mutually orthogonal subspaces:

$$\mathbb{R}^n = \underbrace{\mathcal{C}(\mathbf{X}_0)}_{\dim r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X}_0)^\perp}_{\dim q=r-r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X})^\perp}_{\dim n-r}, \quad \mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M}). \quad (11.2.1)$$

The middle subspace, $\mathcal{C}(\mathbf{M} - \mathbf{M}_0)$, is the **test space** of the hypothesis. It consists of the directions in which the full model can move the mean and the reduced model cannot. Every data vector splits accordingly,

$$\mathbf{y} = \mathbf{M}_0\mathbf{y} + (\mathbf{M} - \mathbf{M}_0)\mathbf{y} + (\mathbf{I} - \mathbf{M})\mathbf{y},$$

and so does the mean vector, $\mathbf{X}\boldsymbol{\beta} = \mathbf{M}_0\mathbf{X}\boldsymbol{\beta} + (\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta} + \mathbf{0}$. The hypothesis says that the middle component of the *mean* is zero. The middle component of the *data*, $(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}$, is its unbiased estimate, with covariance $\sigma^2(\mathbf{M} - \mathbf{M}_0)$. The residual component $(\mathbf{I} - \mathbf{M})\mathbf{Y}$ has mean zero whether or not the hypothesis holds, and it serves as a yardstick for noise. The first component $\mathbf{M}_0\mathbf{Y}$ estimates the part of the mean that both models allow. It carries no information about the hypothesis, and the F statistic ignores it.

What the statistic does use is $(\mathbf{I} - \mathbf{M}_0)\mathbf{y}$, the residual vector of the reduced model. It is split into a piece in the test space and a piece in the residual space, and

$$F = \frac{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2/q}{\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2/(n-r)}$$

compares their squared lengths per dimension. Under the hypothesis both pieces are pure noise, and each dimension carries on average σ^2 . Under the alternative the first piece also carries the signal $(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}$.

The test space can be described by a spanning set. Since $\mathbf{M} - \mathbf{M}_0 = (\mathbf{I} - \mathbf{M}_0)\mathbf{M}$ by Theorem 6.5.1(a), and $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$,

$$\mathcal{C}(\mathbf{M} - \mathbf{M}_0) = \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}).$$

The test space is spanned by the columns of $\mathbf{X}$ after the reduced model has been regressed out of them. When $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_2]$ adds a block $\mathbf{X}_2$, this is $\mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2)$, the space of Lemma 9.2.1. For a hypothesis on coefficients it is the constraint space $\mathcal{C}(\mathbf{M}\mathbf{T})$ of Theorem 6.5.2, as [Section 11.3](#) shows.

11.2.2 One angle

Because the two pieces of $(\mathbf{I} - \mathbf{M}_0)\mathbf{y}$ are orthogonal, their lengths are the legs of a right triangle whose hypotenuse is the reduced model's residual vector.

Proposition 11.2.1. F as a function of an angle

Let $\mathbf{w} = (\mathbf{I} - \mathbf{M}_0)\mathbf{y} \neq \mathbf{0}$ and let $\theta \in [0, \pi/2]$ be the angle between $\mathbf{w}$ and the test space, defined by $\cos \theta = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|/\|\mathbf{w}\|$. Then

$$\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \|\mathbf{w}\|^2 \cos^2 \theta, \quad \text{SSE} = \|\mathbf{w}\|^2 \sin^2 \theta, \quad F = \frac{n-r}{q} \cot^2 \theta.$$

The partial coefficient of determination of Definition 9.5.1 is $\cos^2 \theta$, and the F test rejects exactly when $\theta < \theta_\alpha$, where $\tan^2 \theta_\alpha = (n-r)/\{q F_\alpha(q, n-r)\}$.

Proof. $\mathbf{w} = (\mathbf{M} - \mathbf{M}_0)\mathbf{y} + (\mathbf{I} - \mathbf{M})\mathbf{y}$ with orthogonal summands, since $(\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$. Pythagoras gives $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \|\mathbf{w}\|^2 - \|\mathbf{w}\|^2 \cos^2 \theta = \|\mathbf{w}\|^2 \sin^2 \theta$. Since $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$ is the projection of $\mathbf{w}$ onto the test space, θ is also the angle between $\mathbf{w}$ and its projection, the smallest angle between $\mathbf{w}$ and any vector of the test space. The ratio of the two squared lengths is $\cot^2 \theta$, which gives F . The partial coefficient of determination is $(\text{SSE}_0 - \text{SSE})/\text{SSE}_0 = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2/\|\mathbf{w}\|^2 = \cos^2 \theta$. Finally $\cot^2$ is strictly decreasing on $(0, \pi/2]$, so $F > F_\alpha$ iff $\cot^2 \theta > q F_\alpha/(n-r)$ iff $\theta < \theta_\alpha$. ■

So the F test asks a geometric question: *does the reduced model's residual vector point suspiciously close to the test space?* Under the hypothesis, $\mathbf{w}$ is noise spread over $n - r_0$ dimensions, only q of which belong to the test space, so it typically points well away from it. The rejection region $\{\mathbf{y} : \theta < \theta_\alpha\}$ is a circular cone around the test space inside $\mathcal{C}(\mathbf{X}_0)^\perp$, extended by arbitrary components in $\mathcal{C}(\mathbf{X}_0)$. That this is the same as the partial R^2 test of Theorem 9.5.4(b) is no coincidence. The partial R^2 is a squared cosine, just as Theorem 6.7.2 showed for the ordinary R^2 .

Example. The region test as an angle

In Example (Do regions matter?), the reduced model's residual vector has a component of length 3.851 in the three-dimensional test space and a component of length 9.334 in the 43-dimensional residual space. The angle between the residual vector and the test space is $\theta = 67.6^\circ$, and the partial R^2 is $\cos^2 \theta = 0.145$. The 5% test rejects when $\theta < 66.1^\circ$, so the data fall just outside the cone.

Figure 11.2.1 plots the two lengths for the data and for 4000 simulated data sets. With the reduced model true, 0.053 of them fall in the rejection region. With the Northeast lowered by two murders per 100,000, and all else as fitted, the fraction is 0.774.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statecrime.load_pandas().data
```

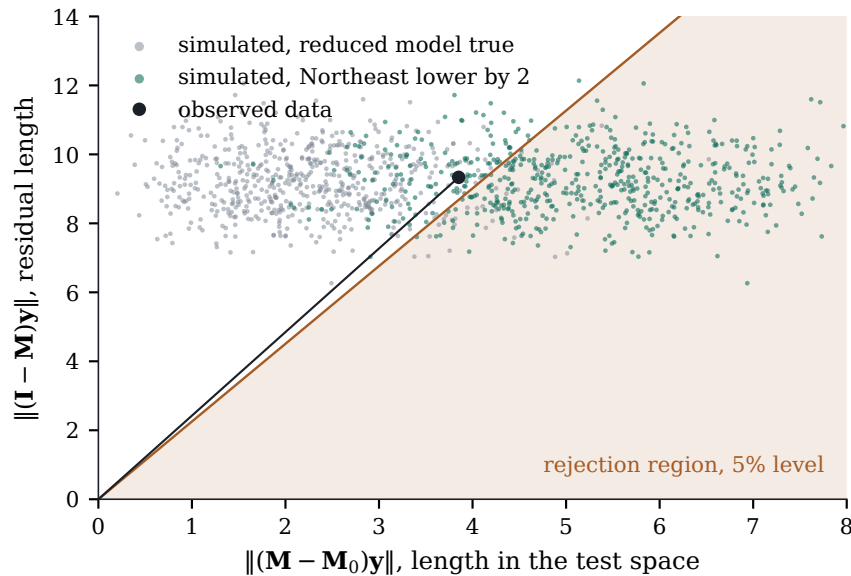


Figure 11.2.1: The region test of Example (Do regions matter?) in two coordinates: the length of $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$ in the test space and the length of the residual vector $(\mathbf{I} - \mathbf{M})\mathbf{y}$. The F statistic depends only on the angle of the point, and the test rejects inside the shaded wedge. The grey points are simulated with the reduced model true, the green points with the Northeast lowered by two murders per 100,000. The observed data are the black point, just above the wedge.

```
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNNMSSSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates])
D = np.column_stack([(region == g).astype(float) for g in "NSW"])
X = np.column_stack([X0, D])

def proj(Z):
    """Orthogonal projection onto  $C(Z)$ , from an orthonormal basis."""
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T

M, M0 = proj(X), proj(X0)
q, nu = 3, n - 7
u = (M - M0) @ y
e = y - M @ y
w = y - M0 @ y
    # r - r0 and n - r
    # component in the test space
    # residual vector
    # what the reduced model leaves
    unexplained
theta = np.arccos(np.linalg.norm(u) / np.linalg.norm(w))
F = (nu / q) / np.tan(theta) ** 2
theta_crit = np.arctan(np.sqrt(nu / (q * stats.f.ppf(0.95, q, nu))))
print(f"angle = {np.degrees(theta):.1f} degrees, F = {F:.3f}")
print(f"reject when the angle is below {np.degrees(theta_crit):.1f} degrees")
```

[Run this code online](#)

11.2.3 What the statistic ignores

Since F depends on $\mathbf{y}$ only through the angle θ , many transformations of the data leave it unchanged. They form the symmetry group of the testing problem, and [Section 11.4](#) uses them to characterize the F test.

Proposition 11.2.2. Invariance of the F statistic

The statistic Eq. 11.1.1 depends on $\mathbf{X}$ and $\mathbf{X}_0$ only through $\mathcal{C}(\mathbf{X})$ and $\mathcal{C}(\mathbf{X}_0)$. Where it is defined, that is, when $(\mathbf{I} - \mathbf{M})\mathbf{y} \neq \mathbf{0}$, it is unchanged when $\mathbf{y}$ is replaced by

$$a\mathbf{H}\mathbf{y} + \mathbf{v}, \quad a \neq 0, \quad \mathbf{v} \in \mathcal{C}(\mathbf{X}_0),$$

where $\mathbf{H}$ is any orthogonal matrix that maps $\mathcal{C}(\mathbf{X}_0)$ onto itself and $\mathcal{C}(\mathbf{X})$ onto itself.

Proof. The first claim holds because $\mathbf{M}$ and $\mathbf{M}_0$ are determined by the column spaces. For the second, the transformations can be applied one at a time. Adding $\mathbf{v}$ changes nothing, because $\mathbf{M} - \mathbf{M}_0$ and $\mathbf{I} - \mathbf{M}$ annihilate $\mathcal{C}(\mathbf{X}_0)$. The factor a^2 cancels in the ratio. For $\mathbf{H}$: the matrix $\mathbf{H}\mathbf{M}\mathbf{H}^\top$ is symmetric and idempotent with column space $\mathbf{H}\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X})$, so it equals $\mathbf{M}$ (Theorem 6.3.3). Hence $\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H}$, and likewise $\mathbf{H}\mathbf{M}_0 = \mathbf{M}_0\mathbf{H}$. Then $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{H}\mathbf{y}\| = \|\mathbf{H}(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\| = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|$, because $\mathbf{H}$ preserves lengths, and the same holds for $\mathbf{I} - \mathbf{M}$. ■

The orthogonal matrices allowed here are exactly those that rotate the test space within itself and the residual space within itself, and fix $\mathcal{C}(\mathbf{X}_0)$ as a set. The invariance says that no direction within the test space is favoured. A departure of given size $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|$ is equally detectable in every direction. When some directions matter more than others, for instance when one contrast is of primary interest, a test directed at them can have more power there. It pays for this with less power elsewhere ([Section 11.6](#)).

11.2.4 Canonical form

Orthonormal bases adapted to [Eq. 11.2.1](#) turn every linear hypothesis into the same problem about independent normal coordinates.

Proposition 11.2.3. Canonical form of a linear hypothesis

Let $\mathbf{Q} = [\mathbf{Q}_0, \mathbf{Q}_1, \mathbf{Q}_2]$ be an orthogonal $n \times n$ matrix whose blocks of r_0 , q and $n - r$ columns are orthonormal bases of $\mathcal{C}(\mathbf{X}_0)$, of the test space and of $\mathcal{C}(\mathbf{X})^\perp$. Put $\mathbf{z}_k = \mathbf{Q}_k^\top \mathbf{Y}$ and $\boldsymbol{\eta}_k = \mathbf{Q}_k^\top \mathbf{X}\boldsymbol{\beta}$, $k = 0, 1, 2$. Under Eq. 7.3.1:

- (a) $\mathbf{z}_0, \mathbf{z}_1, \mathbf{z}_2$ are independent, $\mathbf{z}_k \sim N(\boldsymbol{\eta}_k, \sigma^2 \mathbf{I})$, and $\boldsymbol{\eta}_2 = \mathbf{0}$;
- (b) $\|\mathbf{z}_1\|^2 = \text{SSE}_0 - \text{SSE}$ and $\|\mathbf{z}_2\|^2 = \text{SSE}$, so $F = \{\|\mathbf{z}_1\|^2/q\} / \{\|\mathbf{z}_2\|^2/(n - r)\}$;
- (c) $\|\boldsymbol{\eta}_1\|^2 = \sigma^2 \gamma$, and H_0 holds iff $\boldsymbol{\eta}_1 = \mathbf{0}$.

Proof.

- (a) $\mathbf{Q}^\top \mathbf{Y} \sim N_n(\mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{Q}^\top \mathbf{Q}) = N_n(\mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ by Theorem 3.3.1. Its blocks are uncorrelated, hence independent (Theorem 3.4.1). The columns of $\mathbf{Q}_2$ are orthogonal to $\mathcal{C}(\mathbf{X})$, so $\boldsymbol{\eta}_2 = \mathbf{Q}_2^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{0}$. (b) By Proposition 6.3.2, $\mathbf{Q}_1 \mathbf{Q}_1^\top = \mathbf{M} - \mathbf{M}_0$ and $\mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I} - \mathbf{M}$, so $\|\mathbf{z}_1\|^2 = \mathbf{Y}^\top \mathbf{Q}_1 \mathbf{Q}_1^\top \mathbf{Y} = \mathbf{Y}^\top (\mathbf{M} - \mathbf{M}_0) \mathbf{Y}$ and similarly for $\mathbf{z}_2$. (c) In the same way $\|\boldsymbol{\eta}_1\|^2 = \|(\mathbf{M} - \mathbf{M}_0) \mathbf{X}\boldsymbol{\beta}\|^2$, and Theorem 11.1.1(c) applies. ■

In canonical form the problem is stripped to its essentials. One observes a q -vector $\mathbf{z}_1$, normal with unknown mean $\boldsymbol{\eta}_1$ and covariance $\sigma^2 \mathbf{I}$, and wishes to test $\boldsymbol{\eta}_1 = \mathbf{0}$. An independent vector $\mathbf{z}_2$ of pure noise, of dimension $n - r$, measures σ^2 . The vector $\mathbf{z}_0$ has an unknown mean that neither hypothesis restricts, so it is of no use. Every hypothesis of this chapter, whether about coefficients, contrasts or whole models, is an instance of this problem with particular q and $n - r$.

11.2.5 Beyond normal errors

The angle picture shows why normality is less important for the null distribution than it first appears. Under H_0 , $(\mathbf{I} - \mathbf{M}_0) \mathbf{Y} = (\mathbf{I} - \mathbf{M}_0) \boldsymbol{\varepsilon}$, so F is a function of the error vector alone, and it is a function of its *direction* only: replacing $\boldsymbol{\varepsilon}$ by $c\boldsymbol{\varepsilon}$, $c > 0$, leaves every angle unchanged. Any error law under which the direction of $\boldsymbol{\varepsilon}$ behaves as it does for normal errors therefore gives the same null distribution. We call $\mathbf{U}$ **uniformly distributed on the unit sphere** if it has the distribution of $\mathbf{Z}/\|\mathbf{Z}\|$ for $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$.

Proposition 11.2.4. Spherical errors

Let $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$, where $\boldsymbol{\varepsilon} = R\mathbf{U}$ for a random variable $R > 0$ and a vector $\mathbf{U}$ uniformly distributed on the unit sphere. The variables R and $\mathbf{U}$ need not be independent. Then $F \sim F(r - r_0, n - r)$.

Proof. Write $g(\mathbf{v}) = \{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{v}\|^2/q\} / \{\|(\mathbf{I} - \mathbf{M})\mathbf{v}\|^2/(n - r)\}$ for vectors with $(\mathbf{I} - \mathbf{M})\mathbf{v} \neq \mathbf{0}$. Under H_0 , $(\mathbf{M} - \mathbf{M}_0)\mathbf{Y} = (\mathbf{M} - \mathbf{M}_0)\boldsymbol{\varepsilon}$ and $(\mathbf{I} - \mathbf{M})\mathbf{Y} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$, since both projections annihilate $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$. So $F = g(\boldsymbol{\varepsilon})$. Because $g(c\mathbf{v}) = g(\mathbf{v})$ for $c > 0$, $F = g(R\mathbf{U}) = g(\mathbf{U})$. The law of $g(\mathbf{U})$ is the law of $g(\mathbf{Z}/\|\mathbf{Z}\|) = g(\mathbf{Z})$ with $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$. By Theorem 11.1.1(d) with $\sigma = 1$, that law is $F(r - r_0, n - r)$. (The events $\mathbf{Z} = \mathbf{0}$ and $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$ have probability zero.) ■

The class of error laws covered is large. If $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$ and $S > 0$ is any random scale independent of $\mathbf{Z}$, or not, then $\boldsymbol{\varepsilon} = S\mathbf{Z} = (S\|\mathbf{Z}\|)(\mathbf{Z}/\|\mathbf{Z}\|)$ qualifies. Taking S^{-2} proportional to a $\chi^2(\nu)$ variable gives the spherical multivariate t distribution. Its components are uncorrelated and heavy-tailed but *not* independent: one common scale inflates or shrinks them all. The F test is exactly valid for all such laws. What the proposition does not cover are independent nonnormal errors, whose direction is not uniform. For them the null distribution is only approximately F .

Example. Four error laws

Keep the design of Example (Do regions matter?) and simulate 20000 data sets with the reduced model true, each with errors from one of four laws: independent normal; spherical

t with three degrees of freedom; independent t with three degrees of freedom; independent centred exponential. The spherical errors are uncorrelated, but their sizes move together: the rank correlation between the absolute values of two components is 0.20. The rejection rates of the 5% test are

Errors	normal	spherical t_3	independent t_3	exponential
Rejection rate	0.0502	0.0502	0.0444	0.0474

with Monte Carlo standard error about 0.0015. The first two agree exactly, not just approximately: the spherical errors were made by rescaling the normal ones, and each rescaled data set gives the same F as its normal original. The two independent nonnormal laws give rates slightly below 0.05. For this design, with 43 residual degrees of freedom and no extreme leverage, the independent t_3 errors make the test slightly conservative, the exponential rate is within Monte Carlo error of 0.05, and both are close to the nominal level.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

rng = np.random.default_rng(1103)

data = sm.datasets.statelcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNSMSWNSWSMW"
region = np.array(list(codes))
n = len(data)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates])
D = np.column_stack([(region == g).astype(float) for g in "NSW"])
X = np.column_stack([X0, D])

def proj(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T

M, M0 = proj(X), proj(X0)
q, nu = 3, n - 7
crit = stats.f.ppf(0.95, q, nu)

def F_stat(E):
    """F statistic for each row of E, used as a data vector under the reduced model.
    """
    num = np.sum((E @ (M - M0)) ** 2, axis=1) / q
    den = np.sum((E - E @ M) ** 2, axis=1) / nu
    return num / den

reps = 20000
Z = rng.standard_normal((reps, n))
scale = 1 / np.sqrt(rng.chisquare(3, size=(reps, 1)) / 3) # one scale per data
set
errors = {
    "normal": Z,
    "spherical t3": Z * scale, # dependent, heavy
```

```

    -tailed
    "independent t3": rng.standard_t(3, size=(reps, n)),
    "exponential": rng.exponential(size=(reps, n)) - 1.0,          # skewed
}
for name, E in errors.items():
    print(f"{name:15s} rejection rate {np.mean(F_stat(E) > crit):.4f}")

```

[Run this code online](#)

Remark (Only the null distribution). [Proposition 11.2.4](#) concerns the level of the test. Under an alternative, F depends on the error length R as well as the direction, and the power under spherical t errors differs from the power under normal errors with the same covariance. Nor does the proposition say that the F test is a *good* test for heavy-tailed errors. Its power can be poor compared with robust methods. Those questions belong to [Chapter 19](#) and [Chapter 21](#).

11.2.6 Exercises

11.2.6.1 A. Check your understanding

Exercise. A1

Use [Proposition 11.2.1](#) to express $\cos^2 \theta$ in terms of F , q and $n - r$. Check the value 0.145 of [Example \(The region test as an angle\)](#) from $F = 2.440$, $q = 3$ and $n - r = 43$, and compute θ_α from $F_{0.05}(3, 43) = 2.822$.

11.2.6.2 B. Practice

Exercise. B1

Show that under H_0 the squared cosine $\cos^2 \theta$ of [Proposition 11.2.1](#) has the $\text{Beta}(q/2, (n - r)/2)$ distribution, and that it is independent of $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{Y}\|^2$. What is its expectation?

Solution. By [Proposition 11.2.3](#), $\cos^2 \theta = U/(U + V)$ with $U = \|\mathbf{z}_1\|^2/\sigma^2 \sim \chi^2(q)$ and $V = \|\mathbf{z}_2\|^2/\sigma^2 \sim \chi^2(n - r)$ independent under H_0 , and $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{Y}\|^2 = \sigma^2(U + V)$. Exercise (B4) gives both the Beta law and the independence of $U/(U + V)$ from $U + V$. The mean of $\text{Beta}(a, b)$ is $a/(a + b)$, here $q/(n - r_0)$: under the hypothesis the residual vector of the reduced model puts, on average, a share of its squared length in the test space equal to that space's share of the dimensions.

Exercise. B2

Prove that $\mathcal{C}(\mathbf{M} - \mathbf{M}_0) = \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X})$, and that for $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_2]$ this equals $\mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2)$. Show that $\dim \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2) = \text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_0)$.

Solution. $\mathbf{M} - \mathbf{M}_0 = (\mathbf{I} - \mathbf{M}_0)\mathbf{M}$ by Theorem 6.5.1(a), so $\mathcal{C}(\mathbf{M} - \mathbf{M}_0) \subseteq \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{M}) = (\mathbf{I} - \mathbf{M}_0)\mathcal{C}(\mathbf{M}) = (\mathbf{I} - \mathbf{M}_0)\mathcal{C}(\mathbf{X}) = \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X})$. Conversely each column $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_j = \mathbf{x}_j - \mathbf{M}_0\mathbf{x}_j$ lies in $\mathcal{C}(\mathbf{X})$ and is orthogonal to $\mathcal{C}(\mathbf{X}_0)$, so it lies in $\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X}_0)^\perp = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$. For $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_2]$, $(\mathbf{I} - \mathbf{M}_0)\mathbf{X} = [\mathbf{0}, (\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2]$. The dimension is $\text{rank}(\mathbf{M} - \mathbf{M}_0) = r - r_0$ (Theorem 6.5.1(c)).

Exercise. B3

Which of the following error vectors satisfy the hypothesis of Proposition 11.2.4? (i) $\boldsymbol{\varepsilon} = \sigma\mathbf{Z}$ with σ fixed; (ii) $\boldsymbol{\varepsilon} = S\mathbf{Z}$ with S uniform on $(1, 2)$ and independent of $\mathbf{Z}$; (iii) $\boldsymbol{\varepsilon}$ with independent Laplace components; (iv) $\boldsymbol{\varepsilon} = \mathbf{D}\mathbf{Z}$ with $\mathbf{D}$ a fixed diagonal matrix with unequal entries; (v) $\boldsymbol{\varepsilon} = \mathbf{Z}/\|\mathbf{Z}\|^2$.

11.2.6.3 C. Going deeper

Exercise. C1

In the canonical form, let two data vectors $\mathbf{y}$ and $\mathbf{y}'$ with $\mathbf{z}_2, \mathbf{z}'_2 \neq \mathbf{0}$ have the same value of F . Construct $a > 0$, $\mathbf{v} \in \mathcal{C}(\mathbf{X}_0)$ and an orthogonal $\mathbf{H}$ of the kind allowed in Proposition 11.2.2 such that $\mathbf{y}' = a\mathbf{H}\mathbf{y} + \mathbf{v}$. (So F is a *maximal invariant*: any statistic unchanged by these transformations is a function of F .)

Exercise. C2

Take a one-way layout with two groups of sizes $n_1 = 5$ and $n_2 = 20$, independent normal errors with standard deviations 3 in the small group and 1 in the large one, and equal means. Compute, by simulation or numerical integration, the true level of the nominal 5% F test of equal means. Explain the direction of the error using the angle picture: which way does the unequal variance tilt the residual vector?